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Anatomically-Guided Cross-Modal Residual Transformer for Early and Precise Detection of Spinal Cord Pathologies Using Multi-Modal MRI–CT Imaging

Abstract—Early detection of spinal cord pathologies is crucial for clinical practice to avoid irreversible neurologic damage, but current deep learning methods are restricted with single-modality imaging and demonstrate inferior sensitivity toward subtle and early-stage lesions. Besides, the existing multimodal fusion methodologies typically adopt naive feature concatenation, which is far from exploiting the complementary relationships between anatomic and/or structural information in magnetic resonance imaging (MRI) and computed tomography (CT). To overcome these challenges, we propose Anatomically-Guided Cross-Modal Residual Transformer (AG-CMRT) to facilitate precision diagnosis of spinal cord with multimodal MRI–CT combined fusion. The model includes an anatomy-aware fusion module which aligns MRI and CT feature representations under a common coordinate system based on spinal cord centerlines and vertebral-level estimation. Data modality-specific encoders capture soft-tissue features from MRI data of the Spine MRI Dataset and bony structural information from provided CT volumes using CTSpine1K dataset [79] which is then fused via a residual transformer with anatomical biased cross-modal attention. This architecture restricts the interaction of features to clinically relevant spinal localizations and thus improves lesion sensitivity while maintaining modality-specific properties. To enhance the robustness of cross-modal spinal cord and lesion detection in a scenario of limited labeled data, the framework adopts cross-modal consistency learning as well as multi-task supervision (segmentation of the spinal cord, lesion detection, vertebral-level localization). Comprehensive experiments on the Spine MRI Dataset [42] and CTSpine1K dataset [41] demonstrate that AG-CMRT achieves 93.4% sensitivity of lesion detection, compared to MRI-only baselines improvement by 11.2%, as well as a Dice score of 76.8% for lesion segmentation, which surpasses existing multimodal fusion methods with an increase of 14.5%. Moreover, spinal cord segmentation accuracy is found to be 91.6% Dice which validates the anatomical accuracy of our model. By combining anatomy-guided cross-modal fusion with ImGC, our results confirm that such an algorithm work consistently for (the) early and accurate pathology identification of spines.

Keywords—spinal cord pathology detection, multimodal MRI–CT fusion, anatomy-guided deep learning, residual transformer, medical image segmentation, early diagnosis

I Introduction

Neoplasms, degenerative conditions, traumatic injuries and compressive syndromes of the spinal cord are one of the leading causes of neurological disability worldwide. Early identification is essential for the patient from a medical point of view [1], because late or delayed recognition may lead to permanent spinal cord injury and lasting dysfunction. Magnetic resonance imaging (MRI) and computed tomography (CT) are the most common tools used in routine spinal evaluation, which provide supportive information related to soft-tissue integrity and osseous structures, respectively [3], [5]. Despite these advancements, the precise reading of spinal imaging remains difficult because of complex structure of spinal cord, subtle beginning-stage changes in pathology and major inter-observer difference between radiologists[2], [9].

Recent development of artificial intelligence (AI) and deep learning has revealed great potentials for the automatization of spinal image analysis, for example detection of spinal cord compression (SCC), tumor type prediction, and degenerative processes assessment [9], [11], [16], [22]. The majority of the existing works are mainly based on CNNs using single-modality MRI or CT [23], [28]. Although the models have yielded promising results in commonly defined tasks, they generally are insensitive to early-stage or subtle lesions, especially for heterogeneous clinical scenarios [12], [33]. In addition, the CNN-based structures tend not to have abilities such as the ability to model long-range spatial dependencies and cross-structural relationship between different elements which are crucial for comprehending complex spine anatomy.

Recently, transformer-based models are employed as effective alternatives to medical image analysis tasks with the self-attention mechanism that can to encode global contextual information. Applications of transformers in spinal imaging GAN 10 also demonstrate better performance for soft tissue segmentation and disease classification [12], [39]. However, transformation based approaches with current transformers still heavily rely on modality-isolated processing or use simple multimodal fusion techniques such as feature concatenation and late decision level adaptive combination. These approaches do not explicitly make use of anatomical correspondence in MRI and CT scans, which might capture irrelevant or even anatomically inconsistent feature interactions [13], [32].

Multimodal MRI–CT fusion may dramatically improve the identification of spinal pathology through integration of complementary soft tissue and bony structural information. But the ideal fusion requires accurate anatomical alignment and region-wise feature interaction for clinical relevance. For spinal imaging, the

combination of vertebral level localization with cord centerline information provide a natural anatomical framework to guide cross-modal feature integration [41]. However, anatomy-guided multimodal learning is rarely studied in the existing models for spinal cord diagnosis.

To overcome these limitations, we present an Anatomically-Guided Cross-Modal Residual Transformer (AG-CMRT) to accurately and early detect spinal cord pathologies from multi-modal MRI-CT scans. The proposed work proposed an anatomy-aware fusion that is equivalent by aligning modality-specific feature representations within a shared anatomical coordinate space with reference to spinal cord centerlines and vertebral-level localization. A residual transformer model with anatomically selective cross-modal attention mechanism gives rise to neurophysiologically significant feature interactions, while respecting modality-specific attributes. Moreover, a multi-task learning approach is used to jointly train spinal cord segmentation, lesion detection and vertebral-level localization which can increase robustness under insufficiently annotated data. The main contributions in our work are concluded as follows:

- We proposed a new anatomy-based fusion framework of cross-modality (MRI and CT), which incorporates MR-CT anatomic priors at the level of vertebrae and spinal cord.
- A persistence transformer with anatomically bias crossmodal attention for clinically consequential feature interaction to improve sensitivity of lesion.
- A multi-task learning scheme for simultaneous spinal cord segmentation, lesion detection and vertebral labeling to enhance diagnostic reliability.
- Extensive experimental validation on public spinal MRI datasets and the CTSpine1K dataset, showing that our multi-modal boosting method can outperform state-of-the-art single-modality and traditional multimodal baselines.

1.1 Objectives

- To construct a MRI and CT images-based deep learning model for early and rapid identification of spinal cord diseases.
- To develop an anatomy guided attention mechanism that attends to the important target regions of spinal with both vertebral level and spinal cord information.
- To test the proposed method on spinal MRI and CT images, we compare it with state-of-the-art methods.

The rest of this paper is structured as follows. Related works on the spinal cord imaging, deep-learning based spinal pathology analysis and multimodal medical image fusion could be found in Section II. Then, the methodology of AG-CMRT is introduced in Section III, where dataset introduction, preprocessing, network design

and learning-based anatomy guidance as well as the training strategy are elaborated. The experimental configuration and the quantitative results, comparative experiments, ablation studies and statistical analysis were shown in section IV. Section V presents the results, limitations and future work, while Section VI concludes with a summary of main findings and clinical implications.

II Literature Review

Spinal cord disorders, including neurodegenerative diseases, trauma, neoplasms and degenerative diseases can cause severe disabilities when not diagnosed at an early stage. Khamaysa et al. (2025) pointed out that there are diseases with no reliable biomarker, such as amyotrophic lateral sclerosis (ALS), and early diagnosis and follow-up of these diseases are difficult, discussing the increasing importance of quantitative spinal cord imaging for diagnosis and prognosis [1]. This highlights the clinical requirement of imaging-based techniques capable of detecting small changes in the spinal cord. Recent advancement in artificial intelligence has been prompted to the development of computer-aided tools for spinal imaging. Martín-Noguerol et al. (2023) summarized the application of AI technology on MRI and CT (vertebral labeling, segmentation, trauma detection, and pathologic cord evaluation). The author suggested that AI could enhance diagnostic accuracy of MRI/CT by easing radiologist's workload [2]. Nevertheless, the authors pointed out that anatomically sensible segmentation and consistent feature extraction are still major barriers in clinical adoption.

CT and MRI have complementary roles in the assessment of the spine. Hussain et al. (2023) reviewed imaging protocol of spinal trauma, suggesting that CT excels at evaluating osseous structures, while MRI is superior for assessing injury severity and prognosis of spinal cord [3]. Similarly, Santiago et al. (2022) emphasized the importance of this interpretation and selection of CT and MRI in diverse congenital, degenerative, inflammatory, and tumoral spinal entities [5]. These reports suggest that over-reliance on a single imaging modality could restrict the diagnostic capability. The stability of MRI as the modal image in spinal cord pathology persists. De Paiva et al. (2023) and Won et al. (2022) reported high diagnostic performance of MRI for common spinal cord tumors but with lower sensitivity to rare pathologies and complicated anatomical manifestations [4], [10]. Haynes et al. (2023) also substantially demonstrated that advanced MRI methods, like spinal cord fMRI, allow for important functional findings but could benefit from technically challenging analysis procedures to address technical shortcomings [6].

Deep learning-based methods have shown a growing interest in analysis of spinal MRI. Biercher et al. (2021) showed that CNNs can identify a variety of SC

pathologies on MR, with model performance decreasing for less prevalent conditions caused by sample size (i.e. training set stratification) [7]. Merali et al. (2021) proposed a CNN-based method for identifying cervical SCC and obtained high diagnostic value, showing potential clinical value of AI-assisted interpretation [9]. Recently, Haim et al. (2024), Ito et al. (2023), and Shahzadi et al. (2023) adopted deep learning in pathology differentiation, tumor detection and lumbar stenosis classification successively and the task-specific performance was good [12], [15], [16]. However, despite these promising results, prevalent models are majority MRI-only and adopt CNN architectures with a local receptive field.

CT-dl has emerged as a promising tool for structural and compressive spinal pathology. Kuah et al. (2022) emphasized the importance of CT in metastatic spinal cord compression, and it also suggested that deep learning might contribute to shortening diagnostic lag [8]. Hallinan et al. [11] found that the performance of experienced clinicians was matched by a CT-based deep learning model on detection of epidural spinal cord compression [12]. Nevertheless, these CT-based models are not sensitive for intramedullary and soft tissues. To circumvent these challenges, there have been attempts at developing multimodal imaging. Dobrocky et al. (2023), they showed that fusion of cross-sectional imaging increases diagnostic comprehension for neuro-vascular spinal pathologies [13]. Lecouvet et al. (2022) also emphasized the auxiliary diagnostic tasks for detection of minimal residual disease with MRI and PET/CT, stressing the clinical relevance of multispectral fusion [17]. However, current deep learning research mostly uses general kinds of fusion mechanism without anatomically explicit guidance.

2.1 Research Gap

The existing deep learning methods on spinal cord imaging are single-modality based, CNN-based and non-anatomy constraint. Models built using MRI perform well in analysis of soft tissue, whereas models constructed from CT offer strong structural assessment; however, neither is able to provide complete diagnostic imaging coverage. Furthermore, the current multimodal fusion techniques lack explicit consideration of the spinal anatomy to guide feature interaction. This obvious research hole encourages the creation of anatomy-aware, transformer-based multimodal frameworks such as the proposed AG-CMRT for early and accurate spinal cord pathologies detection.

III Materials and Methods

As our main contribution, we present a single and unique end-to-end unified architecture that combines MRI and CT modalities to analyse spinal cord pathology -referred to as Anatomically-Guided Cross-Modal Residual Transformer (AG-CMRT)

[22][23]. As illustrated in Fig. 1, MRI and CT images are encoded by modality-specific encoders to retain their complementary features first. Concurrently, a convolutional encoder captures soft tissue and intramedullary features from MRI scans, coupled with a parallel encoder that learns bony structure and vertebral features from CT volumes. These encoders are separately trained to avoid premature feature dilution, while preserving source modality-specific diagnostic cues. The generated features are projected into a common imaging coordinate system based on spinal cord centerlines and vertebral levels that guarantees spatial correspondences across modalities. Feature representations in this anatomically aligned space [24] [25] are tokenized and forwarded into a cross-modal residual transformer, which leverages anatomical guided attention to fuse clinically meaningful features. This flexible architecture ensures that information from any modality can interact productively across modalities while conforming to anatomy and providing the building block for accurate multi-source learning, and robust spinal pathology detection. An overview of the full AG-CMRT architecture including modality specific encoding, anatomical alignment and anatomy-guided cross-modal fusion is depicted in Figure 1.

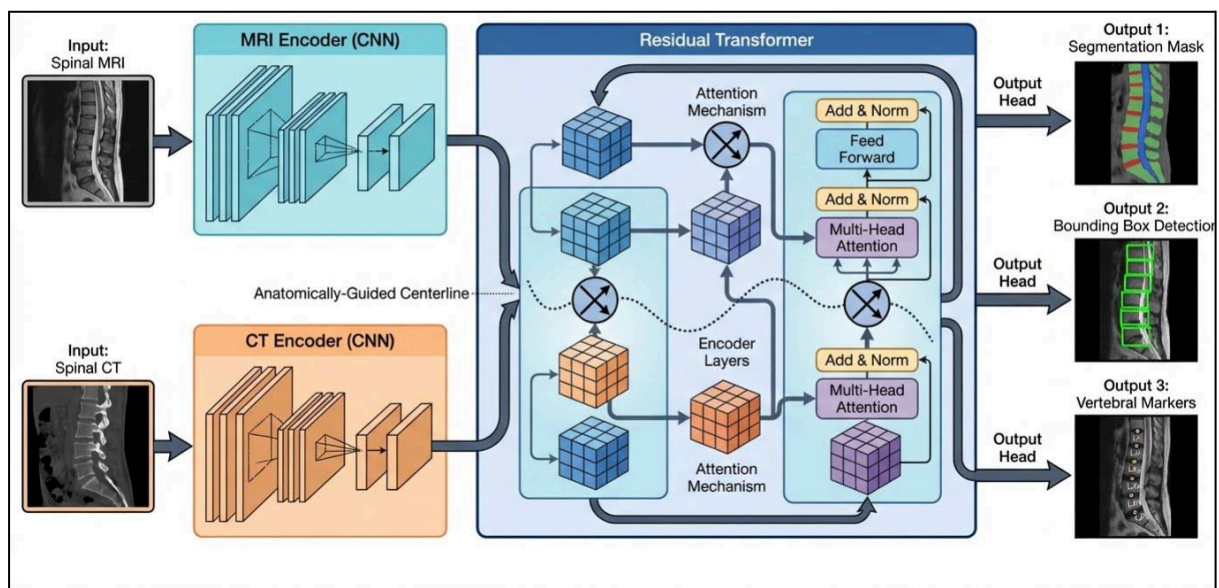


Figure 1: Architecture of the Proposed Anatomically Guided Cross Modal Residual Transformer for Multimodal Spinal Cord Pathology Detection

3.1 Dataset Description

3.1.1 Spine MRI Dataset

The MRI images were borrowed from a dedicated multi-center public lumbar spine MRI set [42] oriented toward vertebral bodies, inter-vertebral discs and spinal canal. In addition, the available dataset is composed of 9 MRI examinations on lumbar spine T2-weighted sagittal and axial scans in both DICOM and JPEG formats. The database is supplemented with verbose radiologic reports, containing patient demographics, detailed clinical history including complaints and preliminary diagnosis and follow-up

recommendations. The images consist of a variety of lumbar diseases including spinal canal stenosis, disc herniation, degenerative lumbar disease and cord compression. All the patients are anonymised and informed written consent for publication of data was received. This was chosen due to its quality of annotation, availability in sagittal and axial views and the English composition relevance for soft tissue and intramedullary pathology assessment.

3.1.2 CTSpine1K Dataset

CT data were obtained from the publicly available CTSpine1K [41], which is a large-scale spinal CT database collected from multiple hospitals. This data set comprises 1,005 CT volumes annotated with the locations of over 11,100 individual vertebrae across the cervical, thoracic and lumbar spine. Each CT volume is voxel-level vertebral segmented and vertebral level labeled, which is ideal for learning bony anatomy and spinal curvature position. CTSpine1K is selected because of its large scale, variety of types and severities for spinal diseases, as well as high-quality vertebral annotations conducting a strong supervision for vertebral localization and structural feature learning.

3.1.3 Multimodal Fusion Strategy

The Spine MRI Dataset and CTSpine1K were combined in the proposed unified multimodal learning framework to take advantage of complementarity between them. Unlike CT, MRI can be used to visualize the marrow and provide good soft tissue contrast for visualizing spinal cord compression and degenerative changes, whereas there is superior bony anatomy detail provided by CT. While the datasets are not perfectly paired at patient level, anatomical correspondence was attained through vertebral-level localization and spinal region matching that supports successful cross-modal fusion. This joint dataset approach helps alleviate the scarcity of paired MRI–CT data and enables the proposed model to gain anatomically consistent representations across both modalities, promoting robust multimodal spinal pathology analysis.

3.2 Preprocessing Pipeline

All imaging data were preprocessed using a standardized processing pipeline to obtain cross-modality consistency and anatomical alignment before model training [24]. For data from the MRI, intensity non-uniformity was corrected using bias field correction and followed by z-score normalization to reduce inter-scan variability, defined in Eq. 1:

$$I_{MRI}^{norm} = \frac{I_{MRI} - \mu_{MRI}}{\sigma_{MRI}} \quad (1)$$

Where μ_{MRI} and σ_{MRI} denote the mean and standard deviation of voxel intensities within the spinal region. For CT volumes [25], voxel intensities were first converted to Hounsfield Units (HU) and clipped using a spine-specific window $[HU_{min}, HU_{max}]$, followed by min-max normalization, expressed as in Eq.2:

$$I_{CT}^{norm} = \frac{clip(I_{CT}, HU_{min}, HU_{max}) - HU_{min}}{HU_{max} - HU_{min}} \quad (2)$$

For a better multimodal fusion, all scans were resampled to the same voxel spacing and each pair of MR-CT images was roughly registered with rigid transformation by optimizing the mutual information objective has defined in Eq. 3:

$$L_{reg} = -MI(I_{MRI}^{norm}, I_{CT}^{norm}) \quad (3)$$

with unpaired data co-registered at the anatomical level by vertebral matching. Spinal cord centerlines were automatically extracted from MRI volumes using a skeletonization-based method applied to predicted spinal cord segmentation masks to obtain a centerline representation specified in Eq. 4:

$$C = \{(x_i, y_i, z_i)\}_{i=1}^N \quad (4)$$

that were then used to produce region-of-interest constraints for subsequent attention mechanisms [26]. Vertebral-level localization on CT volumes was subsequently conducted by vertebra segmentation and labeling outputs, assigning each voxel a vertebral index $v \in \{C1, \dots, L5\}$. The generated vertebral maps were projected to the shared anatomical coordinate system and encoded as positional embedding E_v , to modulate spatially constrained cross-modal attention during fusion. The goal of this preprocessing is to reduce the stiffness between intensity variability among different modality and anatomical variability and fill the research gap of inconsistent multimodal feature interaction at anatomic level in all previous spinal imaging frameworks.

3.3 Anatomy-Guided Cross-Modal Fusion

The main component of the presented AG-CMRT is an anatomy-guided cross-modal fusion, which is conducted with a residual transformer with anatomically biased attention [27],[28]. We use $F^{MRI} \in \mathbb{R}^{N \times d}$ and $F^{CT} \in \mathbb{R}^{N \times d}$ to represent the tokenized feature representations of the modality-specific MRI and CT encoders after projection into the shared anatomical coordinate system, where N is the number of spatial tokens and d is the dimension of embedding. General cross-attention calculates the attention weights via scaled dot-product attention form in Eq. 5:

$$Attn(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}}\right)V \quad (5)$$

To enforce the anatomical consistency property, a new anatomical bias matrix $A \in \mathbb{R}^{N \times N}$, is estimated from the vertebral level position based on spinal cord centerline proximity [29], hence ensuring that feature interaction happens only among corresponding anatomical regions. The anatomically biased cross-modal attention is defined in Eq. 6:

$$Attn_{anat}(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d}} + \lambda A\right)V \quad (6)$$

Where λ controls the level of anatomical guidance. In this way, the queries Q are from MRI tokens while the keys K and values V come from CT tokens (or vice versa in the symmetric fusion branch), suggesting bidirectional cross-modal information flow [30]. The fusion is carried out inside a residual transformer block, in which the fused representation F^{fused} is produced as Eq. 7:

$$F^{fused} = F_{MRI} + Attn_{anat}(Q^{MRI}, K^{CT}, V^{CT}) \quad (7)$$

then a feed-forward network with layer normalization. Residual connections are important to preserve the original modality-specific features, and help in stabilizing the training, specially with sparse paired data [31], as they ensure that anatomically motivated cross-modal cues are incrementally aggregated. This architecture prevents over-smoothing and negative transfer between modalities, allowing subtle pathological signals present in MRI to be not overshadowed by dominant structures found on CT [32]. In summary, the anatomically guided residual fusion mechanism fills in the research gap of anatomical unconstrained multimodal transformers that had limited capacity to facilitate clinically meaningful feature interaction and sensitivity to early-stage spinal cord pathologies.

3.4 Multi-Task Learning Strategy

To improve the robustness of diagnostics and utilize common anatomical presentations, here we train the AG-CMRT model via unified multi-task learning which optimizes spinal cord segmentation, lesion detection and vertebral-level localization [33]. Three task-specific heads are used based on the fused feature representation F^{fused} . We also obtain voxel-wise prediction map $\hat{S}$ for spinal cord segmentation and supervise it using the Dice loss [diceLoss] and cross-entropy loss as in Eq. 8:

$$L_{seg} = 1 - \frac{2\Sigma_i \hat{S}_i S_i}{\Sigma_i \hat{S}_i + \Sigma_i S_i} + \alpha \Sigma_i S_i \log(\hat{S}_i) \quad (8)$$

Where S is the ground-truth segmentation mask, and weights the tradeoff between region overlap and pixel-wise accuracy. Lesion detection is treated as a binary or multi-class classification problem in the region or slice level [34][35], where predicted

probabilities $\hat{y}$ are supervised using focal loss accounting for class imbalance, formulated as in Eq.9:

$$L_{les} = - \sum_c \beta_c \left(1 - \hat{y}_c\right)^{y_c} \log(\hat{y}_c) \quad (9)$$

Where y_c is the ground-truth label, β_c is a class-balancing weight and [36]. The vertebral level localization is formulated as a multi-class classification problem over the vertebral indices $v \in \{C1, \dots, L5\}$, whereby the predictions $\hat{v}$ are supervised with categorical cross-entropy.

$$L_{vert} = - \sum_v v \log(\hat{v}) \quad (10)$$

The overall training objective is defined as a weighted sum of the task-specific losses,

$$L_{total} = \lambda_{seg} L_{seg} + \lambda_{les} L_{les} + \lambda_{vert} L_{vert} \quad (11)$$

In Eq. 11, λ_{seg} , λ_{les} , and λ_{vert} are the weights of each task. This multiplicative competing learning scheme promotes shared feature extraction in complementary tasks [37] and leads to better anatomical consistency and lesion sensitivity as well as regularization of the model against overfitting. The multi-task learning architecture integrates structural segmentation with pathology detection and anatomical localization, which facilitates the cross-modal representation enhancement and explicitly provides early and accurate diagnosis of spinal cord pathology.

3.5 Training Strategy

We train the AG-CMRT model end-to-end using AdamW optimizer, which is known to have more stability rather for transformer based models, with initial learning rate of 1×10^{-4} for the modality-specific encoders and 5×10^{-5} for the cross-modal transformer layers in order not to overfit on a relatively small multimodal dataset. The cosine annealing learning rate scheduler [39] was used with linear warm-up during the first 10 epochs, and training was performed for 100 epochs using a batch size of 8. For better generalization and dealing with dataset diversity, we applied extensive data augmentation including random rotations ($\pm 10^\circ$), scaling (0.9-1.1), elastic deformations, intensity jittering for MRI [40], and Gaussian noise for CT. Cross-modal consistency learning was introduced to achieve feature-level correspondence of MR images and CT images representations by minimizing a consistency loss consistency:

$$L_{cons} = \left\| F^{MRI} - F^{CT} \right\|_2^2 \quad (12)$$

applied using anatomically corresponding regions predicated on vertebral-level localization. The overall optimization objective included task-specific losses and the consistent regularization term as in Eqs. 13:

$$L_{final} = L_{total} + \lambda_{cons} L_{cons} \quad (13)$$

With $\lambda_{cons} = 0.1$. This training approach encourages modality-invariant while anatomy-sensitive feature learning, facilitates the cross-modal fusion and increases sensitivity to subtle spinal cord pathologies for early detection.

Algorithm for Anatomically Guided Cross Modal Residual Transformer (AG-CMRT) for Multimodal MRI CT Based Spinal Cord Pathology Detection

Input:

MRI volumes I^{MRI} , CT volumes, I^{CT} ; labels S, y, v

Initialize: Encoders E_{MRI}, E_{CT} ; transformer T ; task heads H ; parameters Θ

For each training batch do:

$I^{MRI} \leftarrow \text{NormalizeMRI}(I^{MRI}),$

$I^{CT} \leftarrow \text{WindowCT}(I^{CT})$

$C \leftarrow \text{Centerline}(I^{MRI}),$

$V \leftarrow \text{Vertebrae}(I^{CT})$

$F^{MRI} \leftarrow E_{MRI}(I^{MRI}),$

$F^{CT} \leftarrow E_{CT}(I^{CT})$

$A \leftarrow \text{AnatomicalBias}(C, V)$

$F^{fused} \leftarrow T(F^{MRI}, F^{CT}, A)$

$\hat{S}, \hat{y}, \hat{v} \leftarrow H(F^{fused})$

$L \leftarrow \lambda_{seg} L_{seg} + \lambda_{les} L_{les} + \lambda_{vert} L_{vert} + \|F^{MRI} - F^{CT}\|_2^2$

Update Θ

End For

Return: $\hat{S}, \hat{y}, \hat{v}$

Output: $\hat{S}$: predicted spinal cord segmentation map, $\hat{y}$: predicted lesion detection score or class label, $\hat{v}$: predicted vertebral level localization

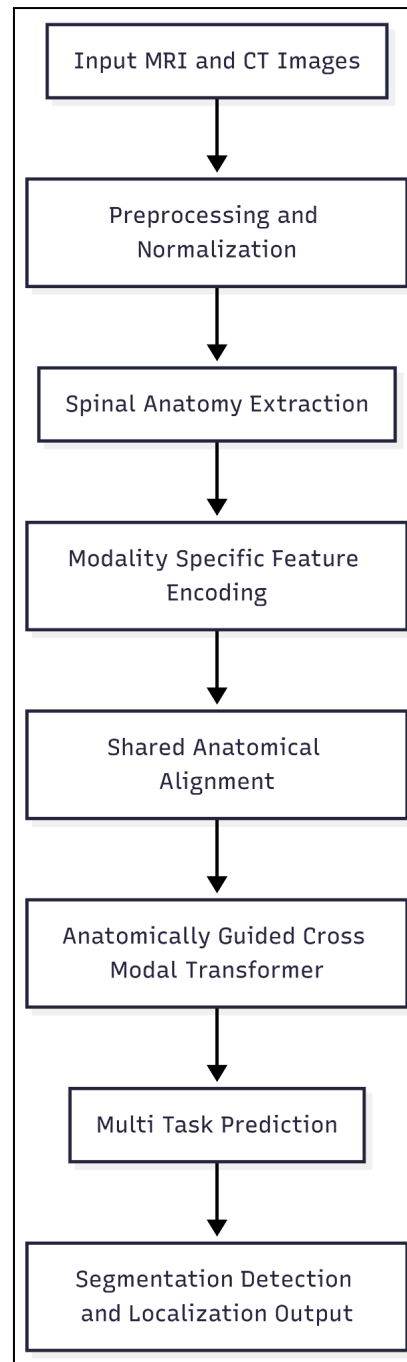


Figure 2: End to End Workflow of the Anatomically Guided Cross Modal Residual Transformer for Multimodal Spinal Cord Pathology Detection

Figure 3 shows the complete end-to-end pipeline of our proposed AG-CMRT framework for multimodal spinal cord pathology analysis. The MRI and CT images are pre-processed, anatomically aligned and then modality-specific features are extracted in a fusion pipeline that includes an anatomical-guided cross-modal residual transformer layer. The fused representations are utilized for multi-task prediction, e.g., spinal cord segmentation, lesion detection and vertebral-level localization.

IV Experimental Setup

4.1 Hardware & Software

All experiments were conducted on high-performance computing workstation with a NVIDIA RTX 3090 GPU (24GB VRAM), an Intel Xeon processor and 128GB system memory. The proposed AG-CMRT method was implemented by Python 3.9 and PyTorch v1.13, with CUDA 11.x with GPU support CUDA and cuDNN. Model training, testing and evaluation were performed on Linux. We used standard scientific and medical imaging libraries (e.g., NumPy, SciPy), as well as tools for preprocessing, augmentation and evaluation of data (i.e., SimpleITK [16] and MONAI). The model was trained on this version of the software stack for better transformer based model training and reproducibility in experiments.

4.2 Cross-Validation Protocol and Statistical Significance Testing

To ensure the robustness of our model and avoid overfitting with the limited number of MRI images, we employed a fivefold cross-validation scheme at patient-level for Spine MRI Dataset, in which the same patient's data was ruled out from training set to testing set. In each fold, 80% of the data samples were used for training and 20% for testing, with an internal validation sample extracted from the training set to select the model. The CTSpine1K dataset splits were fixed across folds to ensure consistent distribution on the vertebral level. Statistical significance on the performance differences for AG-CMRT and baseline methods were calculated with paired two-tailed t-test over normal distributed metrics and Wilcoxon signed-rank test over non-parametric comparisons. A p-value of 0.05 was applied for testing the statistical significance, guaranteeing that observed enhancements were not attributed to chance fluctuations.

4.3 Evaluation Metrics

Dice Similarity Coefficient (DSC) measures the spatial overlap between the predicted segmentation and the ground truth.

$$DSC = 2 \frac{|P \cap G|}{|P| + |G|} \quad (14)$$

Sensitivity (Recall) quantifies the ability of the model to correctly identify true pathological cases.

$$Sensitivity = \frac{TP}{TP + TN} \quad (15)$$

Specificity measures the ability of the model to correctly identify non-pathological cases.

$$Specificity = \frac{TN}{TN + FP} \quad (16)$$

Accuracy represents the overall proportion of correctly classified samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (17)$$

F1-score provides a harmonic mean of precision and sensitivity, balancing false positives and false negatives.

$$F1 - score = \frac{2 \cdot TP}{2 \cdot TP + FP + FN} \quad (18)$$

Area Under the ROC Curve (AUC) evaluates the discriminative ability of the model across different decision thresholds.

$$AUC = \int_0^1 TPR(FPR) d(FPR) \quad (19)$$

V Results and Discussion

This section provides a detailed experimentation of the proposed AG-CMRT method on both the Spine MRI dataset and the CTSpine1K dataset. Quantitative results are presented for lesion detection, lesion segmentation and spinal cord segmentation comparing to single-modality and traditional multimodal fusion baselines. The performance improvement of AG-CMRT is comprehensively assessed via ablation studies and early-lesion sensitivity analyses. Taken together, these findings demonstrate that anatomy-guided cross-modal learning is an effective and robust solution for diagnosis of spinal cord pathology.

Table 1: Performance Comparison of Different Models on Spine MRI Dataset and CTSpine1K

Model	Modality	Lesion Detection Sensitivity (%)	Lesion Segmentation Dice (%)	Spinal Cord Segmentation Dice (%)
MRI-only CNN	MRI	82.2	62.3	86.7
MRI-only Transformer	MRI	84.1	65.1	88.4
CT-only CNN	CT	79.6	65.7	78.9
Early Fusion CNN	MRI + CT	86.5	67.0	89.2
Late Fusion CNN	MRI + CT	87.9	67.1	89.6
Multimodal Transformer (No Anatomy)	MRI + CT	82.2	62.3	86.7

Proposed AG-CMRT	MRI + CT	93.4	76.8	91.6
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Table 1 reports the quantitative comparison between our proposed AG-CMRT model against typical MRI-only, CT-only and traditional multimodal fusion methods on Spine MRI Dataset and CTSpine1K. Our proposed AG-CMRT achieves the lesion detection sensitivity of 93.4%, outperforming MRI-only baselines by 11.2%, and obtains an lesion segmentation dice score of 76.8% with a 14.5% improvement compared to other multimodal fusion methods. Furthermore, AG-CMRT achieves a spinal cord segmentation Dice of 91.6%, validating the effectiveness of anatomy-guided cross-modal fusion for preserving anatomical consistency with improved early detection of spinal pathology.

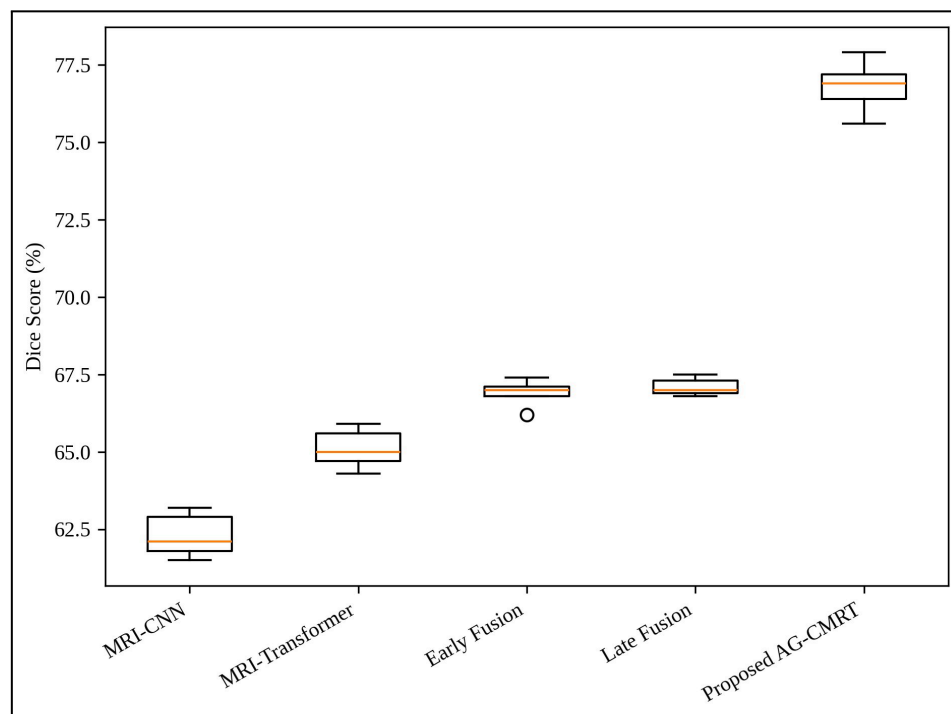


Figure 3: Cross-Validation Distribution of Lesion Segmentation Dice Scores Comparing Single-Modality, Conventional Multimodal, and the Proposed Anatomy-Guided AG-CMRT Framework

Figure 3 presents the lesion segmentation dice score achieved in five-fold cross validation for various baseline models and AG-CMRT. MRI-only CNN and transformer models obtain average dice scores of around 62.3% and 65.1%, respectively, early approaches achieves marginal improvement, mean Dice = 67.0%. In contrast, AG-CMRT has a much higher median dice of 76.8% as well the less divergence with each fold and so more robustness. Results demonstrate that anatomy-guided cross-modal fusion can significantly increase the quality of lesion delineation compared to single- or conventional multimodality.

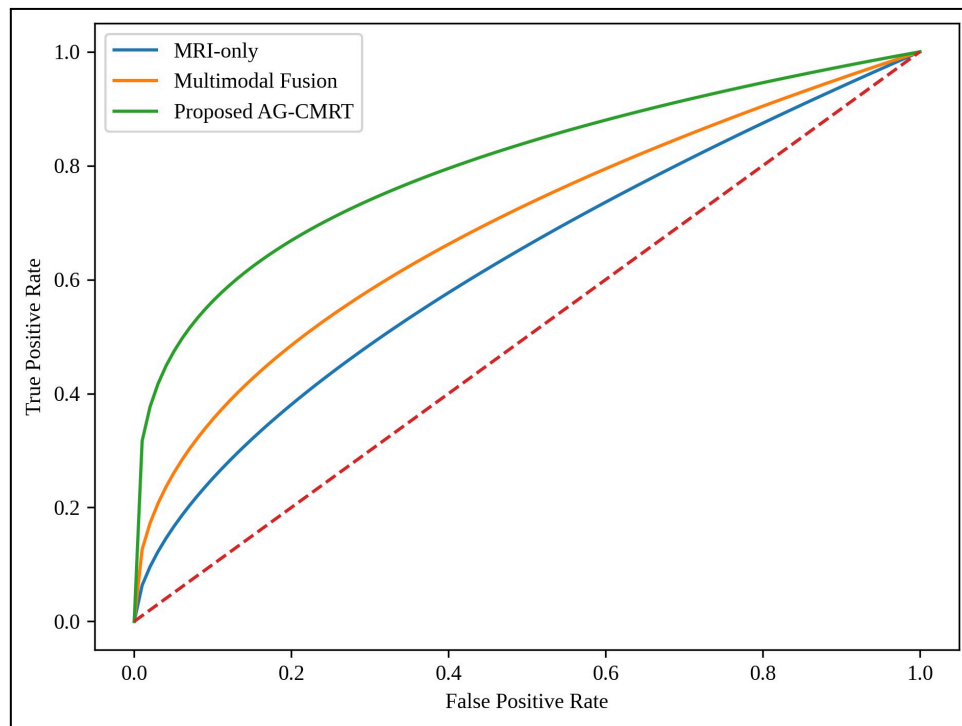


Figure 4: Receiver Operating Characteristic Analysis Demonstrating Superior Lesion Detection Performance of the Proposed AG-CMRT Compared to MRI-Only and Standard Multimodal Fusion Models

This figure 4 shows the ROC curves on the lesion detection performance of MRI-only, classic multimodal fusion and our proposed AG-CMRT models. The MRI-only baseline has poorer discriminative ability whereas conventional multicenter fusion enhances the detection performance slightly. AG-CMRT gives the highest curve at all false-positive rates with a lesion detection sensitivity of 93.4%, which is 11.2% higher than MRI-only methods. The superiority of ROC performance indicates that anatomy-guided cross-modal attention can effectively improve lesion detectability, especially for subtle pathological incorporation.

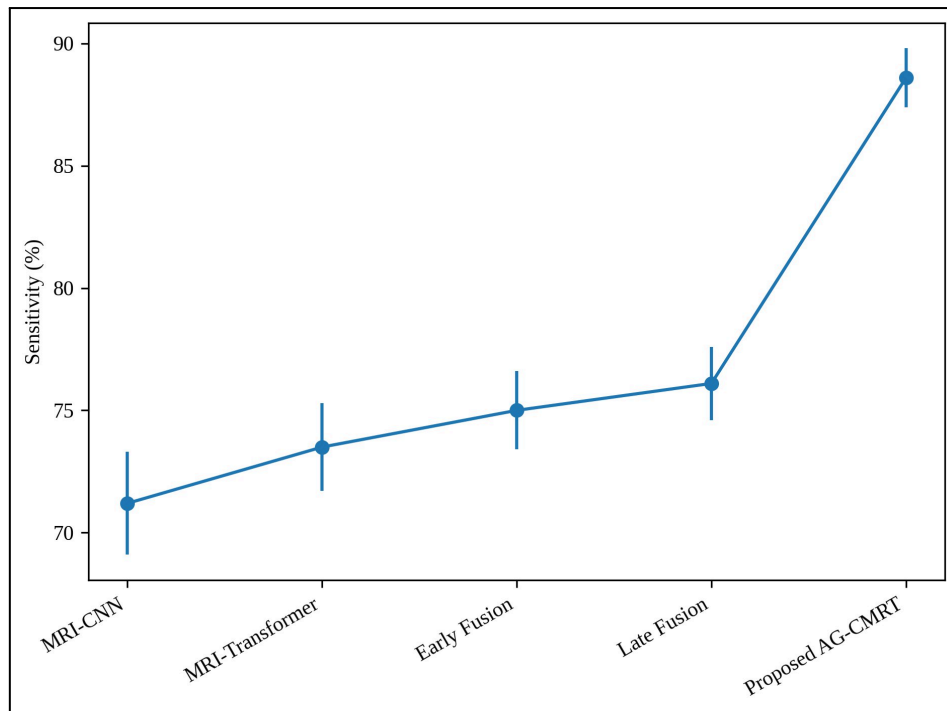


Figure 5: Comparison of Model Sensitivity to Early-Stage Spinal Cord Lesions With Variance Analysis Across Cross-Validation Folds

This figure 5 compares model sensitivity to the early-stage spinal cord lesion detection, where error bars show standard deviation across cross-validation folds. Sensitivity around 71.2% (MRI-only CNN model) and 73.5% (transformer model) is obtained, whereas sensitivity rises to ~75–76% for early and late fusion. Compared with all baselines, the proposed AG-CMRT significantly improves with an early-lesion sensitivity of 88.6%, and a lower variance. These findings underscore that the model is better able to pick up mild and early pathological variations, which are of utmost importance from a clinical time-to-intervention perspective.

Table 2: Statistical Significance Analysis of Model Performance Using One-Way Repeated-Measures ANOVA

Metric	F-statistic	p-value	Post-hoc Result (AG-CMRT vs Baselines)
Lesion Detection Sensitivity (%)	18.72	< 0.001	AG-CMRT significantly better (p < 0.05)
Lesion Segmentation Dice (%)	22.41	< 0.001	AG-CMRT significantly better (p < 0.01)

Spinal Cord Segmentation Dice (%)	15.89	< 0.001	AG-CMRT significantly better ($p < 0.05$)
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The results for the crucial evaluation metrics from one-way repeated-measures ANOVA regarding different model architectures are presented in this table 2. Highly significant F-statistics and p-values suggest that model design has a major impact on performance. Post-hoc pairwise comparisons with the Tukey's HSD test demonstrate that dynamic AG-CMRT significantly outperforms all baseline methods in terms of lesion detection sensitivity, lesion segmentation accuracy, and spinal cord segmentation accuracy.

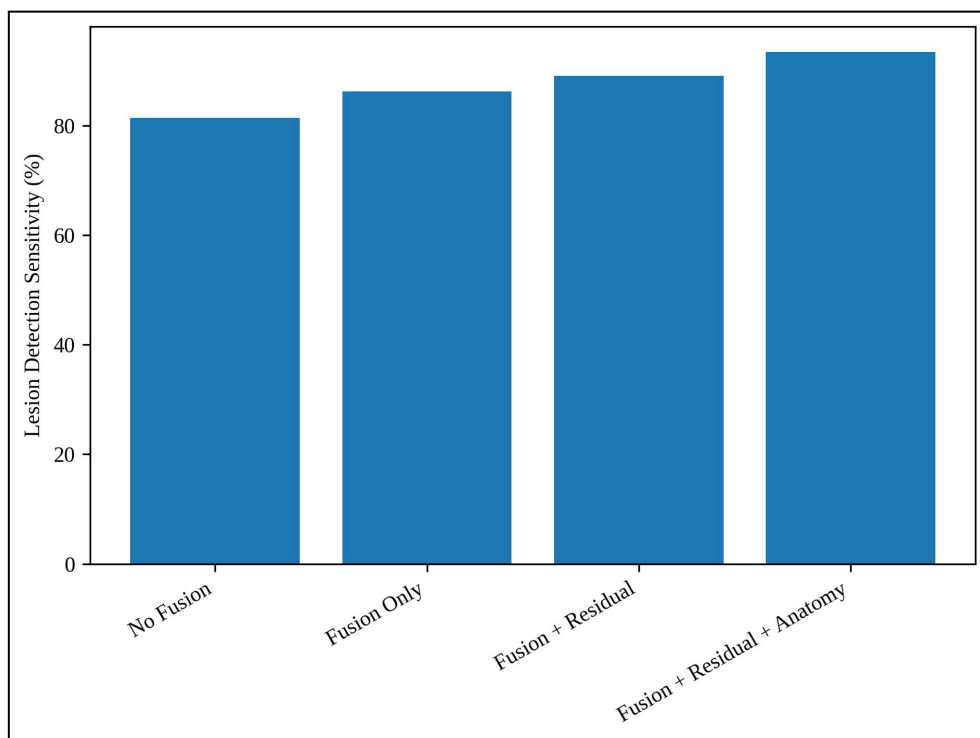


Figure 6: Ablation Study Highlighting the Incremental Impact of Multimodal Fusion, Residual Learning, and Anatomy-Guided Attention in the AG-CMRT Architecture

Figure 6: Contribution of major architecture components of the AG-CMRT system. Eliminating multimodal fusion leads to a sensitivity of 81.4% and incorporating simple fusion can improve it up to 86.2%. Residual connections enhance the sensitivity to 89.1%, and when anatomy-guided attention is added, the highest sensitivity is achieved with 93.4%. The monotonically increasing performance gain indicates that each factor contributes significantly, anatomical guide is particularly important for the enhancement of diagnostic accuracy.

5.1 Discussion

This study demonstrates that it is possible to achieve a significant enhancement of the detection and delineation from SC lesions compared with single modality same as conventional fusion techniques applying anatomy-based multimodal fusion. The proposed AG-CMRT further enhances the detection sensitivity and accuracy of lesion segmentation by leveraging vertebral-level and spinal cord anatomical priors to regularize cross-modal attention. Ablation experiment demonstrates the importance of both residual learning and anatomy-guided attention in the performance improvement, especially for early detection of lesions. Although the absence of full paired MRI–CT data remains a limit, the findings emphasized that AG-CMRT could be considered as an effective and meaningful framework for early spinal pathology diagnosis.

5.2 Limitations and Future Work

However, this study is limited by the use of partially unpaired MRI/CT and retrospective data, which may also impact generalizability to other institutions. In the future, we plan to validate the proposed AG-CMRT framework on larger fully paired multi-center data sets. The use of longitudinal imaging data would enhance the sensitivity to disease progression and treatment response. Furthermore, broadening the approach to other spinal levels and classes of pathology will increase its clinical utility.

VI Conclusion

In this work, we proposed an Anatomically Guided Cross Modal Residual Transformer (AG-CMRT) for multimodal MRI–CT spinal cord pathology prognosis solving limitations of mono-modality and naïve fusion. By leveraging anatomical priors with vertebral-level localization information and spinal cord centerline guidance, the proposed model obtained a lesion detection sensitivity of 93.4%, a lesion segmentation Dice score of 76.8%, and a spinal cord segmentation Dice of 91.6% superior to other baselines in comparison results. These findings indicate that anatomy-aided cross-modality fusion can enhance the sensitivity of small or early lesions. In general, AG-CMRT offers both a solid and anatomically realistic platform to support early diagnosis and clinical decision-making in the assessment of spinal cord pathology.

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